20 m CW Transmitter — xmitter

KiCad Symbol Library Reference

Library file: **xmitter_prj/kicad/xmitter.kicad_sym**

Generated: **2026-06-04**

Symbols: **8 components defined**

Project: **20 m CW transmitter, solid-state VFO + digital keyer + vacuum-tube PA**

Symbol	Description	Package	Manufacturer	MPN
MC1496	Balanced Modulator / Demodulat…	DIP-14	onsemi	MC1496P
MCP4921	12-bit SPI DAC, Single Channel	DIP-8	Microchip Technology	MCP4921-E/P
TL072	Dual Low-Noise JFET Op-Amp	DIP-8 (3…	Texas Instruments	TL072CP
ADS1115	16-bit 4-Channel I²C ADC	MSOP-10	Texas Instruments	ADS1115IDGST
ST7735_Display	1.8″ SPI TFT Display Breakout	10-pin header	Adafruit Industries	Adafruit #498
W5500	Hardwired TCP/IP Ethernet Cont…	8-pin header	WIZnet	W5500
6N137	High-Speed Optoisolator (10 MB…	DIP-8	Lite-On	6N137
Encoder_5734	60 mm Rugged Rotary Encoder, 1…	Panel mount, screw terminals	Adafruit Industries	Adafruit #5734

Note: Symbols are hand-created from datasheet pinouts. Verify footprint assignments before PCB layout.

MC1496 — Balanced Modulator / Demodulator

Package: DIP-14 Manufacturer: onsemi MPN: MC1496P

#	Name	Function	#	Name	Function
1	**SIG_IN+**	Signal input + (AGC/modula…	8	**CAR_IN-**	Carrier input −
2	**GADJ_A**	Gain adjust A	9	**NC**	—
3	**GADJ_B**	Gain adjust B	10	**CAR_IN+**	Carrier input +
4	**SIG_IN-**	Signal input − (AGC/modula…	11	**NC**	—
5	**BIAS**	Output bias	12	**OUT-**	Collector output −
6	**OUT+**	Collector output +	13	**NC**	—
7	**NC**	—	14	**VEE**	Positive supply (+12 V)

Design notes & gotchas:

- VEE (pin 14) is the POSITIVE supply — counterintuitive Motorola naming convention.
- SIG_IN pins (1, 4) take the 0–10 V key-shaping voltage in this design (AGC/modulating port).
- Pins 7, 9, 11, 13 are NC — leave unconnected; no X markers required.
- MC1496P (DIP) has limited availability. SOIC MC1496DR2G is more readily stocked.

MCP4921 — 12-bit SPI DAC, Single Channel

Package: DIP-8 Manufacturer: Microchip Technology MPN: MCP4921-E/P

#	Name	Function	#	Name	Function
1	**VDD**	Positive supply (3.3 V)	5	**~LDAC**	Latch DAC output, active low
2	**~CS**	SPI chip select, active low	6	**VREF**	Reference voltage input
3	**SCK**	SPI clock	7	**VSS**	Ground
4	**SDI**	SPI data input	8	**VOUT**	DAC analog output (0 – VREF)

Design notes & gotchas:

- Connect VREF to 3.3 V rail — output range is 0–3.3 V before the TL072 gain stage (×3 → 0–9.9 V).
- ~LDAC: pull permanently low for immediate output update on each SPI write.
- SPI command word: 0x3000 OR 12-bit data (gain=1×, unbuffered, active mode).
- Shares SPI bus with ST7735 and W5500 — each device must have its own CS line.

TL072 — Dual Low-Noise JFET Op-Amp (3 units: A, B, Power)

Package: DIP-8 Manufacturer: Texas Instruments MPN: TL072CP

#	Name	Function	#	Name	Function
1	**OUT [A]**	Op-amp 1 output	5	**IN+ [B]**	Op-amp 2 non-inverting input
2	**IN− [A]**	Op-amp 1 inverting input	6	**IN− [B]**	Op-amp 2 inverting input
3	**IN+ [A]**	Op-amp 1 non-inverting input	7	**OUT [B]**	Op-amp 2 output
4	**VCC−**	Negative supply (−8.2 V fr…	8	**VCC+**	Positive supply (+12 V fro…

Design notes & gotchas:

- Multi-unit symbol: place U?A and U?B as separate op-amp instances; U?C for power connections.
- Supply: +12 V / −8.2 V from existing rig rails — 20.2 V total, within 36 V absolute max.
- Output swings to within ~2 V of +rail: 10 V output is achievable with a +12 V supply.
- Tie unused unit: IN+ to GND via 10 kΩ, IN− to its own output. Never leave inputs floating.

ADS1115 — 16-bit 4-Channel I^2C ADC

Package: MSOP-10   Manufacturer: Texas Instruments   MPN: ADS1115IDGST

#	Name	Function	#	Name	Function
1	**ADDR**	I^2C address select	6	**AIN2**	Analog input 2
2	**~ALERT/RDY**	Alert / conversion ready, …	7	**AIN3**	Analog input 3
3	**GND**	Ground	8	**VDD**	Positive supply (2.0–5.5 V)
4	**AIN0**	Analog input 0	9	**SDA**	I^2C data, bidirectional
5	**AIN1**	Analog input 1	10	**SCL**	I^2C clock

Design notes & gotchas:

- ADDR sets I^2C address: GND→0x48, VDD→0x49, SDA→0x4A, SCL→0x4B.
- ~ALERT/RDY is open-drain — pull-up resistor required (4.7 kΩ to VDD).
- ADS1115IDGST is tape-and-reel (2500 min). Verify single-piece orderable PN at Digi-Key.
- MSOP-10 is 0.5 mm pitch — use a breakout board for breadboard prototyping.
- Selected over the ESP32-S3 onboard ADC, which is notoriously nonlinear.

ST7735_Display — 1.8″ SPI TFT Display Breakout (Adafruit #498)

Package: 10-pin header   Manufacturer: Adafruit Industries   MPN: Adafruit #498

#	Name	Function	#	Name	Function
1	**GND**	Ground	6	**~TFT_CS**	TFT chip select, active low
2	**VCC**	Supply (3.3–5 V)	7	**MOSI**	SPI data in
3	**~RESET**	Reset, active low	8	**SCK**	SPI clock
4	**D/C**	Data / Command select	9	**MISO**	SPI data out (SD card only)
5	**~CARD_CS**	SD card chip select, activ…	10	**LITE**	Backlight LED anode

Design notes & gotchas:

- Pin numbers match the Adafruit #498 header order — verify against board before PCB layout.
- ~CARD_CS and MISO only needed if using the onboard SD card; tie ~CARD_CS high if unused.
- LITE can be PWM-controlled for brightness, or tied directly to 3.3 V for always-on backlight.
- Shares SPI bus with MCP4921 and W5500 — separate CS lines required for each.
- Will be panel-mounted externally with a laser-cut acrylic bezel (SendCutSend).

W5500 — Hardwired TCP/IP Ethernet Controller Module

Package: 8-pin header   Manufacturer: WIZnet   MPN: W5500

#	Name	Function	#	Name	Function
1	**~INT**	Interrupt, active low, ope…	5	**MISO**	SPI data out
2	**~CS**	SPI chip select, active low	6	**~RST**	Reset, active low
3	**SCK**	SPI clock	7	**3V3**	Positive supply (3.3 V)
4	**MOSI**	SPI data in	8	**GND**	Ground

Design notes & gotchas:

- Currently backordered — all other subsystems can be developed and tested without it.
- Pin order varies by module vendor (Adafruit, WIZnet EVB, clones) — verify when module arrives.
- ~INT is open-drain — pull-up resistor required to 3.3 V.
- ~RST should be asserted briefly on power-up; drive from MCU GPIO or an RC reset circuit.
- Must be enclosed inside the RFI shielded box alongside the Metro ESP32-S3.

6N137 — High-Speed Optoisolator (10 MBd, open-collector)

Package: DIP-8   Manufacturer: Lite-On   MPN: 6N137

#	Name	Function	#	Name	Function
1	**A**	LED anode	5	**VO**	Output, open-collector
2	**K**	LED cathode	6	**VE**	Enable, active high
3	**NC**	—	7	**VCC**	Output-side supply
4	**GND**	Output-side ground	8	**NC**	—

Design notes & gotchas:

⚠ ⚠ VE (pin 6, Enable) MUST be tied to VCC — leaving it floating causes erratic or no output.
- VO is open-collector — external pull-up resistor required (typ. 470 Ω to VCC).
- Two isolated power domains: LED side (pins 1–2) and logic side (pins 4, 7). Keep grounds separate.
- LED series resistor required on input: 270 Ω for ~10 mA at 5 V, 150 Ω at 3.3 V.
- One 6N137 per paddle input — use two devices for dit and dah isolation.

Encoder_5734 — 60 mm Rugged Rotary Encoder, 100 PPR (Adafruit #5734)

Package: Panel-mount, screw terminals   Manufacturer: Adafruit Industries   MPN: Adafruit #5734

#	Name	Function
1	**A**	Quadrature output A
2	**B**	Quadrature output B
3	**C**	Common (ground reference)

Design notes & gotchas:

- C (common) is the ground reference — connect to MCU ground.
- A and B need pull-up resistors (10 kΩ) or enable MCU internal pull-ups in firmware.
- 100 PPR = 400 counts/revolution with quadrature (×4) decoding in software.
- If rotation direction is reversed, swap A and B in firmware — no hardware change needed.

⚠ ⚠ Poor manufacturer documentation — verify screw terminal labeling with ohmmeter before wiring.
- Footprint is a placeholder 3-pin header; replace with screw-terminal connector for PCB.